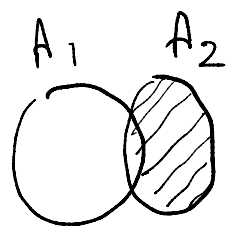


PRST Class 15/09/2020

Result $P\left(\bigcup_{i=1}^K A_i\right) \leq \sum_{i=1}^K P(A_i)$

proof: $A_1 \cup A_2 = \underbrace{A_1} \cup \underbrace{A_2 - A_1}_{A_2 - A_1 \cap A_2^c}$

$$P(A_1 \cup A_2) = P(A_1) + P(A_2 - A_1)$$
$$\leq P(A_1) + P(A_2)$$



Use induction.

Conditional Probability:

$$\Omega = \{ \overset{\checkmark}{1} \overset{\checkmark}{2} \overset{\checkmark}{3} \overset{\checkmark}{4} \overset{\checkmark}{5} \overset{\checkmark}{6} \overset{\checkmark}{7} \overset{\checkmark}{8} \overset{\checkmark}{9} \overset{\checkmark}{0} \} \quad 10$$

$$B = \{ \overset{\frac{1}{7}}{2} \ 3 \ 4 \ 5 \ 6 \ 8 \ 9 \} \quad A = 2 \text{ occurs}$$

$$P(2 | B) = \frac{1}{7} = \frac{\frac{1}{10}}{\frac{7}{10}} = \frac{1}{7}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(\Omega) = 1$$

A and B are independent events

$$P(A|B) = P(A)$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = P(A) \Rightarrow P(A \cap B) = P(A) \cdot P(B)$$

If the sequence of events $\{B_i\}$ in Ω is not mutually exclusive (disjoint) and

$$\bigcup_i B_i = \Omega$$

then $\{B_i\}$ forms a partition of the sample space.

Result: Theorem of total probability.

$$\begin{aligned} P(A) &= P(A \cap \Omega) \\ &= P(A \cap \bigcup_i B_i) \\ &= P(\bigcup_i \underbrace{A \cap B_i}) \\ &= \sum_i P(A \cap B_i) \\ &= \sum_i P(B_i) P(A|B_i) \end{aligned}$$

Bayes' theorem:

$\{B_i\}$ forms a partition of Ω .
A is any event of Ω .

A is any event

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_i P(B_i)P(A|B_i)}$$

proof

$$P(B_i|A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(B_i)P(A|B_i)}{P(A)}$$

Ex. In a MCQ there are 3 options
(only 1 option is correct).
If a student knows the right answer
then the prob of giving the right ans is 1.
If he does not know, the prob of giving
the correct ans is $\frac{1}{3}$.

If a student got his answer right,
what is the prob that he knows the
ans?

C - give correct ans

K - knows the ans

C^c

NK - does not know the ans

$$\begin{array}{|l} P(K) = p \\ P(NK) = 1-p \end{array}$$

$$P(K|C) = ?$$

$$= \frac{P(K)P(C|K)}{P(K)P(C|K) + P(NK)P(C|NK)}$$

$$= \frac{p}{p + (1-p)\frac{1}{3}}$$